41107 Marine and Ocean Engineering

FLOW AROUND AND FORCES ON CYLINDERS

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects all reviewed questions beginning with **FLOW AROUND AND FORCES ON CYLINDERS**, plus the directly related **Morison / Froude-Krylov / diffraction** questions that belong to the same physical block. Exact duplicates across the reviewed MCQ documents were removed. The questions are ordered as requested: first **steady current**, then **oscillatory flow / wave-related cylinder questions**, and finally **Froude-Krylov / diffraction / large-body questions**.
For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants, unless stated otherwise

$$\rho = 1025 \text{ kg/m}^3, \qquad \nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}, \qquad g = 9.81 \text{ m/s}^2$$

$$A = \frac{\pi D^2}{4}, \qquad \omega = \frac{2\pi}{T}$$

Contents

1 Steady current 2

2 Oscillatory flow / wave-related FLOW-labelled questions 12

3 Froude-Krylov / diffraction / large-body questions 24

4 Final formula reminder for this group 26

Steady current

S1. Question 1

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder: The drag force can be split into a form drag and a friction drag. Which is largest when $Re > 10^4$?

Options:

A. friction drag
B. equal magnitude
C. form drag

Correct answer: C. form drag

Formulario / principles used

- For a bluff body such as a circular cylinder at offshore-relevant Reynolds numbers, drag is mainly pressure/form drag.
- Mean drag can be decomposed conceptually as

$$F_D = F_{\text{form}} + F_{\text{friction}}.$$

- For $Re \gtrsim 10^4$, the wake and pressure difference dominate over wall friction.

Step-by-step solution

1. Identify the condition:
$$Re > 10^4.$$
No extra numerical calculation is needed because the Reynolds-number threshold is already given.
2. At this Reynolds number, separation creates a large wake behind the cylinder.
3. The main contribution to drag is therefore the pressure difference between the front and rear of the cylinder:
$$F_{\text{form}} \gg F_{\text{friction}}.$$
4. Hence the largest contribution is form drag.

Source(s): Test MC exam_Corr / Test_Exam_all

S2. Question 2

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder. How many vortices are shed during one vortex shedding period?

Options:

A. 1
B. 4
C. 2

Correct answer: C. 2

Formulario / principles used

- Vortex shedding behind a free circular cylinder is alternating: one vortex from one side, then one from the other side.
- One complete shedding period contains one complete alternating cycle.

Step-by-step solution

1. During one half of the cycle, one vortex is shed from one side of the cylinder.
2. During the other half of the cycle, one vortex is shed from the opposite side.
3. Therefore, over one full vortex shedding period:

$$N_{\text{vortices}} = 1 + 1 = 2.$$

Source(s): Test MC exam_Corr / Test_Exam_all

S3. Question 3

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder: What is the pressure in excess of hydrostatic pressure in the stagnation point?

Options:

A. $c_p = 0$
B. $\frac{1}{2}\rho U_c^2$
C. 1 Pa

Correct answer: B. $\frac{1}{2}\rho U_c^2$

Formulario / principles used

- Stagnation pressure increase from Bernoulli:

$$p_s - p_0 = \frac{1}{2}\rho U_c^2.$$

- At the stagnation point, the local velocity is zero and the incoming kinetic energy per unit volume becomes pressure.

Step-by-step solution

1. Apply Bernoulli between the undisturbed flow and the stagnation point:

$$p_s - p_0 = \frac{1}{2}\rho U_c^2.$$

2. No numerical value of U_c is given, so the result remains symbolic.
3. Therefore, the pressure in excess of hydrostatic pressure is

$$p_s - p_0 = \frac{1}{2}\rho U_c^2.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

S4. Question 4

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free smooth circular cylinder, the wake is turbulent, $U_c = 0.5$ m/s, and $D = 0.2$ m. Which flow regime describes the flow?

Options:

A. Trans-critical
B. Super-critical
C. Sub-critical

Correct answer: C. Sub-critical

Formulario / principles used

- Reynolds number:
$$Re = \frac{U_c D}{\nu}.$$
- For a smooth circular cylinder, a useful course-level classification is:
$$300 \lesssim Re \lesssim 3 \times 10^5 \quad \Rightarrow \quad \text{sub-critical regime.}$$

Step-by-step solution

1. Insert the given values into the Reynolds number:
$$Re = \frac{U_c D}{\nu} = \frac{(0.5)(0.2)}{1.0 \times 10^{-6}}.$$
2. Compute the numerator:
$$(0.5)(0.2) = 0.10.$$
3. Divide by the kinematic viscosity:
$$Re = \frac{0.10}{1.0 \times 10^{-6}} = 1.0 \times 10^5.$$
4. Compare with the regime limits:
$$1.0 \times 10^5 < 3 \times 10^5.$$
5. Therefore, the flow is in the sub-critical regime.

Source(s): Test MC exam_Corr / Test_Exam_all

S5. Question 5

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder: Consider the following three cases defined by D and U_c. You can assume $C_D = 1.2$ in the sub-critical flow regime, 0.6 in the super-critical flow regime and 0.9 in the trans-critical flow regime. Which case has the highest drag force?

Options:

A. $D = 0.2$ m, $U_c = 0.5$ m/s
B. $D = 5.0$ m, $U_c = 0.2$ m/s
C. $D = 0.5$ m, $U_c = 0.2$ m/s

Correct answer: B. $D = 5.0$ m, $U_c = 0.2$ m/s

Formulario / principles used

- Mean drag force per unit length:
$$F_D = \frac{1}{2}\rho C_D D U_c^2.$$
- Since ρ and $1/2$ are common to all cases, compare:
$$C_D D U_c^2.$$
- Reynolds number for identifying the regime:
$$Re = \frac{U_c D}{\nu}.$$

Step-by-step solution

1. Case A Reynolds number:
$$Re_A = \frac{(0.5)(0.2)}{1.0 \times 10^{-6}} = 1.0 \times 10^5.$$
This is sub-critical, so $C_D = 1.2$. Drag factor:
$$C_D D U_c^2 = (1.2)(0.2)(0.5)^2 = (1.2)(0.2)(0.25) = 0.060.$$
2. Case B Reynolds number:
$$Re_B = \frac{(0.2)(5.0)}{1.0 \times 10^{-6}} = 1.0 \times 10^6.$$
This is super-critical, so $C_D = 0.6$. Drag factor:
$$C_D D U_c^2 = (0.6)(5.0)(0.2)^2 = (0.6)(5.0)(0.04) = 0.120.$$
3. Case C Reynolds number:
$$Re_C = \frac{(0.2)(0.5)}{1.0 \times 10^{-6}} = 1.0 \times 10^5.$$
This is sub-critical, so $C_D = 1.2$. Drag factor:
$$C_D D U_c^2 = (1.2)(0.5)(0.2)^2 = (1.2)(0.5)(0.04) = 0.024.$$
4. Compare:
$$0.120 > 0.060 > 0.024.$$
5. Therefore, Case B has the highest drag force.

Source(s): MCQ Test 2025 / Test_Exam_all

S6. Question 6

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free smooth circular cylinder, the wake is turbulent and the vortex shedding frequency, $f_v = 0.018$ Hz, $U_c = 0.2$ m/s, and $D = 5$ m. Which flow regime describes the flow?

Options:

A. Trans-critical
B. Super-critical
C. Sub-critical

Correct answer: B. Super-critical

Formulario / principles used

- Reynolds number:
$$Re = \frac{U_c D}{\nu}.$$
- Strouhal number:
$$St = \frac{f_v D}{U_c}.$$
- For a smooth cylinder, $St \approx 0.45$ is typical of the super-critical regime.

Step-by-step solution

1. Compute the Reynolds number:
$$Re = \frac{(0.2)(5)}{1.0 \times 10^{-6}} = \frac{1.0}{1.0 \times 10^{-6}} = 1.0 \times 10^6.$$
2. Compute the Strouhal number:
$$St = \frac{f_v D}{U_c} = \frac{(0.018)(5)}{0.2}.$$
3. Numerical substitution:
$$(0.018)(5) = 0.090, \qquad St = \frac{0.090}{0.2} = 0.45.$$
4. A Strouhal number around 0.45, together with $Re = 10^6$, indicates the super-critical regime.

Source(s): MCQ Test 2025 / Test_Exam_all

S7. Question 7

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. f_v is the vortex shedding frequency. What is the frequency of the oscillating drag force?

Options:

A. $\frac{1}{2} f_v$
B. f_v
C. $2 f_v$

Correct answer: C. $2f_v$

Formulario / principles used

- Oscillating drag-force frequency:
$$f_{D,\text{osc}} = 2f_v.$$
- Physical reason: drag receives two similar pulses per full vortex shedding cycle, while lift changes sign once per alternating cycle.

Step-by-step solution

1. The formula is directly:
$$f_{D,\text{osc}} = 2f_v.$$
2. No numerical value of f_v is given, so the result remains symbolic.
3. Therefore, the oscillating drag force has frequency $2f_v$.

Source(s): MCQ Test 2025 / Test_Exam_all

S8. Question 8

FLOW AROUND AND FORCES ON CYLINDERS. Steady current around a free smooth circular cylinder. Consider a cylinder with $D = 1.0$ m in a current $U_c = 0.25$ m/s. Take $\nu = 1.0 \times 10^{-6}$ m^2/s. Which flow regime is most appropriate?

Options:

A. Sub-critical
B. Critical / trans-critical onset
C. Super-critical
D. No vortex shedding because $Re < 1$

Correct answer: A. Sub-critical

Formulario / principles used

- Reynolds number:
$$Re = \frac{U_c D}{\nu}.$$
- Smooth-cylinder sub-critical range, approximately:
$$300 \lesssim Re \lesssim 3 \times 10^5.$$

Step-by-step solution

1. Substitute the data:
$$Re = \frac{(0.25)(1.0)}{1.0 \times 10^{-6}}.$$
2. Compute:
$$Re = \frac{0.25}{1.0 \times 10^{-6}} = 2.5 \times 10^5.$$
3. Compare with the threshold:
$$2.5 \times 10^5 < 3.0 \times 10^5.$$
4. The value is close to the transition region, but it is still below the usual critical onset.
5. Therefore, the most appropriate answer is sub-critical.

Source(s): 41107_mock_exam_with_answers

S9. Question 9

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. Consider the following three cases. Assume $C_D = 1.2$ in the sub-critical regime and $C_D = 0.9$ in the trans-critical regime. Which case gives the largest mean drag force per unit length?

Options:

A. $D = 0.3$ m, $U_c = 0.4$ m/s
B. $D = 4.0$ m, $U_c = 0.25$ m/s
C. $D = 0.8$ m, $U_c = 0.6$ m/s

Correct answer: C. $D = 0.8$ m, $U_c = 0.6$ m/s

Formulario / principles used

- Mean drag force per unit length:
$$F_D = \frac{1}{2}\rho C_D D U_c^2.$$
- Compare only the proportional factor:
$$F_D \propto C_D D U_c^2.$$

Step-by-step solution

1. Case A:
$$Re_A = \frac{(0.4)(0.3)}{1.0 \times 10^{-6}} = 1.2 \times 10^5.$$
Use $C_D = 1.2$. Then:
$$C_D D U_c^2 = (1.2)(0.3)(0.4)^2 = (1.2)(0.3)(0.16) = 0.0576.$$

2. Case B:
$$Re_B = \frac{(0.25)(4.0)}{1.0 \times 10^{-6}} = 1.0 \times 10^6.$$
Using the high-Re coefficient specified by the question, $C_D = 0.9$:
$$C_D D U_c^2 = (0.9)(4.0)(0.25)^2 = (0.9)(4.0)(0.0625) = 0.225.$$

3. Case C:
$$Re_C = \frac{(0.6)(0.8)}{1.0 \times 10^{-6}} = 4.8 \times 10^5.$$
Using the high-Re coefficient specified by the question, $C_D = 0.9$:
$$C_D D U_c^2 = (0.9)(0.8)(0.6)^2 = (0.9)(0.8)(0.36) = 0.2592.$$

4. Compare the three drag factors:
$$0.2592 > 0.225 > 0.0576.$$

5. Therefore, Case C gives the largest drag force.

Source(s): 41107_mock_exam_with_answers

S10. Question 10

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder. What is the pressure in excess of hydrostatic pressure at the stagnation point if $U_c = 1.6$ m/s and $\rho = 1025$ kg/m^3?

Options:

A. 0 Pa
B. 820 Pa
C. 1310 Pa
D. 2620 Pa

Correct answer: C. 1310 Pa

Formulario / principles used

- Stagnation pressure increase:
$$p_s - p_0 = \frac{1}{2}\rho U_c^2.$$

Step-by-step solution

1. Substitute the data:
$$p_s - p_0 = \frac{1}{2}(1025)(1.6)^2.$$
2. Square the velocity:
$$(1.6)^2 = 2.56.$$
3. Multiply:
$$p_s - p_0 = 512.5 \times 2.56 = 1312 \text{ Pa}.$$
4. Round to the closest option:
$$p_s - p_0 \approx 1310 \text{ Pa}.$$

Source(s): 41107_mock_exam_with_answers

S11. Question 11

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. If the vortex shedding frequency is $f_v = 0.24$ Hz, what is the frequency of the oscillating drag force?

Options:

A. 0.12 Hz
B. 0.24 Hz
C. 0.48 Hz
D. 0.96 Hz

Correct answer: C. 0.48 Hz

Formulario / principles used

- Oscillating drag-force frequency:
$$f_{D,\text{osc}} = 2f_v.$$

Step-by-step solution

1. Substitute the vortex shedding frequency:
$$f_{D,\text{osc}} = 2(0.24 \text{ Hz}).$$
2. Compute:
$$f_{D,\text{osc}} = 0.48 \text{ Hz}.$$

Source(s): 41107_mock_exam_with_answers

S12. Question 12

FLOW AROUND AND FORCES ON CYLINDERS. For a smooth circular cylinder in the sub-critical regime, why is C_D almost constant over a wide range of Re?

Options:

A. Because separation occurs early and at nearly the same angular position, so the wake width stays similar.

B. Because friction drag dominates completely and pressure drag is negligible.
C. Because the boundary layer is turbulent from the stagnation point.
D. Because vortex shedding stops in the sub-critical regime.

Correct answer: A. Because separation occurs early and at nearly the same angular position, so the wake width stays similar.

Formulario / principles used

- Drag coefficient:
$$C_D = \frac{F_D}{\frac{1}{2}\rho U_c^2 D}.$$
- In the sub-critical regime, the boundary layer is laminar up to separation, and the separation point stays relatively stable.

Step-by-step solution

1. In the sub-critical regime, separation occurs early on the cylinder.
2. Because the separation point does not move dramatically with Re, the wake width remains similar.
3. Since form drag is controlled mainly by wake width and pressure difference, C_D remains approximately constant.
4. No numerical substitution is required because this is a conceptual regime question.

Source(s): 41107_mock_exam_with_answers

Oscillatory flow / wave-related FLOW-labelled questions

O1. Question 1

FLOW AROUND AND FORCES ON CYLINDERS. The wave period is T. What is the definition of the KC number?

Options:

A. $KC = \dfrac{U_m T}{D}$

B. $KC = \dfrac{U_m D}{T}$

C. $KC = \dfrac{TD}{U_m}$

Correct answer: A. $KC = \dfrac{U_m T}{D}$

Formulario / principles used

- Keulegan-Carpenter number:
$$KC = \frac{U_m T}{D}.$$
- $U_m T$ has units of length and represents a characteristic fluid excursion over one period.

Step-by-step solution

1. Start from the definition:
$$KC = \frac{\text{fluid excursion length}}{\text{cylinder diameter}}.$$
2. The characteristic excursion length is:
$$\text{fluid excursion} \sim U_m T.$$
3. Therefore:
$$KC = \frac{U_m T}{D}.$$
4. No numerical substitution is needed because the question asks for the definition.

Source(s): Test MC exam_Corr

O2. Question 2

FLOW AROUND AND FORCES ON CYLINDERS. The wave period is T. What is the definition of the KC number?

Options:

A. $KC = \dfrac{U_m D}{\nu}$

B. $KC = \dfrac{0.44}{D/L}$

C. $KC = \dfrac{U_m T}{D}$

Correct answer: C. $KC = \frac{U_m T}{D}$

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

- Do not confuse it with wave Reynolds number:

$$Re = \frac{U_m D}{\nu}.$$

Step-by-step solution

1. Option A, $U_m D/\nu$, is the wave Reynolds number, not KC.
2. Option B is not the definition of KC.
3. Option C matches the correct definition:

$$KC = \frac{U_m T}{D}.$$

4. No numerical substitution is needed because the question asks for the definition.

Source(s): MCQ Test 2025 / Test_Exam_all

O3. Question 3

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free, smooth circular cylinder: $D = 1$ m, $T = 10$ s, $U_m = 0.4$ m/s. What is the maximum value of the Froude-Krylov force?

Options:

A. 396 N/m
B. 193 N/m
C. 202 N/m

Correct answer: C. 202 N/m

Formulario / principles used

- Froude-Krylov force per unit length:

$$F_{FK} = \rho A \dot{U}.$$

- For $U(t) = U_m \sin(\omega t)$, the maximum acceleration is:

$$\dot{U}_{\max} = U_m \omega.$$

- Therefore:

$$F_{FK,\max} = \rho A U_m \omega.$$

- Circular cylinder cross-sectional area:

$$A = \frac{\pi D^2}{4}.$$

- Angular frequency:

$$\omega = \frac{2\pi}{T}.$$

Step-by-step solution

1. Compute the cross-sectional area:

$$A = \frac{\pi D^2}{4} = \frac{\pi (1)^2}{4} = 0.7854 \text{ m}^2.$$

2. Compute the angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{10} = 0.6283 \text{ rad/s}.$$

3. Substitute into the maximum Froude-Krylov force:

$$F_{FK,\max} = \rho A U_m \omega = (1025)(0.7854)(0.4)(0.6283).$$

4. Numerical multiplication:

$$F_{FK,\max} = 202.3 \text{ N/m}.$$

5. Closest option:

$$F_{FK,\max} \approx 202 \text{ N/m}.$$

Source(s): Test MC exam_Corr / Test_Exam_all

O4. Question 4

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free, smooth circular cylinder. Consider the case defined by $U_m = 1$ m/s, $D = 1$ m and the wave period $T = 5$ s. Further, assume the following functional expression for determining C_D and C_M. $C_D = 1.3 + 0.1(KC - 12)$ and $C_M = \min(2.0; 2.0 - 0.044(KC - 3))$ for $2 < KC < 12$. What is the ratio between the maximum inertia and maximum drag force ($F_{I,\max}/F_{D,\max}$)?

Options:

A. 0.16
B. 4.00
C. 6.29

Correct answer: C. 6.29

Formulario / principles used

- Keulegan-Carpenter number:
$$KC = \frac{U_m T}{D}.$$
- Maximum inertia-to-drag ratio:
$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{KC}\frac{C_M}{C_D}.$$
- Coefficients from the question:
$$C_D = 1.3 + 0.1(KC - 12), \qquad C_M = \min\left(2.0,\ 2.0 - 0.044(KC - 3)\right).$$

Step-by-step solution

1. Compute KC:
$$KC = \frac{(1)(5)}{1} = 5.$$
2. Compute C_D:
$$C_D = 1.3 + 0.1(5 - 12) = 1.3 + 0.1(-7) = 1.3 - 0.7 = 0.6.$$
3. Compute C_M:
$$C_M = \min\left(2.0,\ 2.0 - 0.044(5 - 3)\right).$$
$$C_M = \min(2.0,\ 2.0 - 0.088) = \min(2.0,\ 1.912) = 1.912.$$
4. Substitute into the force ratio:
$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{5}\frac{1.912}{0.6}.$$
5. Numerical calculation:
$$\frac{\pi^2}{5} = \frac{9.8696}{5} = 1.9739, \qquad \frac{1.912}{0.6} = 3.1867.$$
$$\frac{F_{I,\max}}{F_{D,\max}} = (1.9739)(3.1867) = 6.29.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

O5. Question 5

MORISON EQUATION - Froude-Krylov vs Inertia Force. What is the relationship between the Froude-Krylov force F_K and the total inertia force F_I in the Morison equation?

Options:

A. $F_I = C_M F_K$ (the Froude-Krylov is multiplied by C_M)
B. $F_I = F_K / C_M$

C. $F_I = F_K +$ extra diffraction force

Correct answer: A. $F_I = C_M F_K$

Formulario / principles used

- Morison equation for a stationary cylinder in oscillatory flow:

$$F(t) = \frac{1}{2}\rho C_D D U|U| + \rho C_M A\dot{U}.$$

- Froude-Krylov force:

$$F_K = \rho A\dot{U}.$$

- Total inertia force:

$$F_I = \rho C_M A\dot{U}.$$

- Coefficient relation:

$$C_M = C_m + 1.$$

Step-by-step solution

1. Start from the Froude-Krylov force:

$$F_K = \rho A\dot{U}.$$

2. Start from the inertia term in the Morison equation:

$$F_I = \rho C_M A\dot{U}.$$

3. Factor out the Froude-Krylov part:

$$F_I = C_M(\rho A\dot{U}).$$

4. Since $\rho A\dot{U} = F_K$:

$$F_I = C_M F_K.$$

5. Therefore, the total inertia force is the Froude-Krylov force multiplied by C_M.

Source(s): 41107_MCQ_Study_Guide

O6. Question 6

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free circular cylinder. $KC < 2$ and $Re \gg 10^3$ therefore the flow around the cylinder can be described as

Options:

A. A pair of symmetric vortices
B. No separation
C. Vortex shedding

Correct answer: B. No separation

Formulario / principles used

- In oscillatory flow, the Keulegan-Carpenter number controls the fluid excursion relative to the diameter:
$$KC = \frac{U_m T}{D}.$$
- Very small KC means the fluid does not travel far enough around the cylinder for separation and vortex shedding to develop.

Step-by-step solution

1. The key given condition is:
$$KC < 2.$$
2. This means the fluid excursion is less than about two diameters.
3. Even though $Re \gg 10^3$, the oscillatory excursion is too small for a developed wake to form.
4. Therefore, the best description is:
$$\text{No separation.}$$
5. No numerical substitution is needed because the question gives the regime directly through $KC < 2$.

Source(s): Test MC exam_Corr / Test_Exam_all

O7. Question 7

FLOW AROUND AND FORCES ON CYLINDERS. A circular cylinder is placed on the seabed. $D = 0.5$ m, water depth $h = 12$ m, wave height $H = 1$ m and wave period $T = 7$ s. What is the KC number?

Options:

A. $KC = 6.3$
B. $KC = 4.2$
C. $KC = 5.5$

Correct answer: B. $KC = 4.2$

Formulario / principles used

- Keulegan-Carpenter number:
$$KC = \frac{U_m T}{D}.$$
- For a cylinder at the seabed, use the maximum horizontal orbital velocity at $z = -h$:
$$U_m = \frac{\pi H}{T \sinh(kh)}.$$
- Wave dispersion relation:
$$\omega^2 = gk \tanh(kh), \qquad \omega = \frac{2\pi}{T}.$$

Step-by-step solution

1. Compute angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{7} = 0.8976 \text{ rad/s}.$$

2. Solve the dispersion relation:

$$\omega^2 = gk\tanh(kh).$$

Numerically:

$$(0.8976)^2 = 9.81k\tanh(12k).$$

This gives:

$$k \approx 0.09897 \text{ m}^{-1}.$$

3. Compute kh:

$$kh = (0.09897)(12) = 1.1876.$$

4. Compute the hyperbolic sine term:

$$\sinh(kh) = \sinh(1.1876) \approx 1.489.$$

5. Compute maximum seabed orbital velocity:

$$U_m = \frac{\pi H}{T\sinh(kh)} = \frac{\pi(1)}{(7)(1.489)}.$$

$$U_m = \frac{3.1416}{10.423} = 0.3015 \text{ m/s}.$$

6. Compute KC:

$$KC = \frac{U_m T}{D} = \frac{(0.3015)(7)}{0.5} = 4.22.$$

7. Closest answer:

$$KC \approx 4.2.$$

Source(s): MCQ Test 2025 / Test_Exam_all

O8. Question 8

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. The normalized fundamental lift frequency is $N_L = 3$. Which name is given to this flow regime?

Options:

A. three pairs, vortex shedding regime
B. double pair, vortex shedding regime
C. single pair, vortex shedding regime

Correct answer: B. double pair, vortex shedding regime

Formulario / principles used

- Normalized fundamental lift frequency:

$$N_L = \frac{f_L}{f_w}.$$

- Course convention for oscillatory-flow regimes:

$$N_L \approx 2 \Rightarrow \text{single-pair regime,}$$

$$N_L \approx 3 \Rightarrow \text{double-pair regime,}$$

$$N_L \approx 4 \Rightarrow \text{three-pair regime.}$$

Step-by-step solution

1. Given:

$$N_L = 3.$$

2. Match this value with the regime map:

$$N_L = 3 \Rightarrow \text{double-pair vortex shedding regime.}$$

3. No further numerical calculation is needed.

Source(s): MCQ Test 2025 / Test_Exam_all

O9. Question 9

FLOW AROUND AND FORCES ON CYLINDERS. What is the maximum wave height on a horizontal bed given a wave period $T = 7$ s and a water depth $h = 19$ m?

Options:

A. $H_{\text{max}} = 15.2$ m
B. $H_{\text{max}} = 9.3$ m
C. $H_{\text{max}} = 4.1$ m

Correct answer: B. $H_{\text{max}} = 9.3$ m

Formulario / principles used

- Wave angular frequency:
$$\omega = \frac{2\pi}{T}.$$
- Linear dispersion relation:
$$\omega^2 = gk\tanh(kh).$$
- Wavelength:
$$L = \frac{2\pi}{k}.$$
- Miche-type breaking limit on a horizontal bed:
$$\frac{H_{\text{max}}}{L} \approx 0.142\tanh(kh),$$
so
$$H_{\text{max}} \approx 0.142L\tanh(kh).$$

Step-by-step solution

1. Compute angular frequency:
$$\omega = \frac{2\pi}{T} = \frac{2\pi}{7} = 0.8976 \text{ rad/s}.$$
2. Solve dispersion:
$$(0.8976)^2 = 9.81k\tanh(19k).$$
This gives:
$$k \approx 0.0881 \text{ m}^{-1}.$$
3. Compute wavelength:
$$L = \frac{2\pi}{k} = \frac{2\pi}{0.0881} = 71.3 \text{ m}.$$
4. Compute kh:
$$kh = (0.0881)(19) = 1.674.$$
5. Compute $\tanh(kh)$:
$$\tanh(1.674) = 0.932.$$
6. Insert into the breaking limit:
$$H_{\text{max}} = 0.142L\tanh(kh) = 0.142(71.3)(0.932).$$
7. Numerical result:
$$H_{\text{max}} = 9.44 \text{ m}.$$
8. Closest option:
$$H_{\text{max}} \approx 9.3 \text{ m}.$$

Source(s): Test MC exam_Corr / Test_Exam_all

O10. Question 10

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder is described by $U(t) = U_m \sin(\omega t)$ with $U_m = 0.9$ m/s, $T = 6.0$ s and $D = 0.60$ m. What is the Keulegan-Carpenter number?

Options:

A. $KC = 6.0$
B. $KC = 9.0$
C. $KC = 12.0$
D. $KC = 15.0$

Correct answer: B. $KC = 9.0$

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

Step-by-step solution

1. Substitute the given values:

$$KC = \frac{(0.9)(6.0)}{0.60}.$$

2. Compute the numerator:

$$(0.9)(6.0) = 5.4.$$

3. Divide by the diameter:

$$KC = \frac{5.4}{0.60} = 9.0.$$

Source(s): 41107_mock_exam_with_answers

O11. Question 11

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. For $U_m = 0.8$ m/s, $D = 0.50$ m and $T = 4.0$ s, assume $C_D = 1.0$ and $C_M = 1.8$. What is the ratio $F_{I,\max}/F_{D,\max}$ approximately?

Options:

A. 0.28
B. 1.39
C. 2.78
D. 5.56

Correct answer: C. 2.78

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

- Maximum inertia-to-drag ratio:

$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{KC}\frac{C_M}{C_D}.$$

Step-by-step solution

1. Compute KC:

$$KC = \frac{(0.8)(4.0)}{0.50} = \frac{3.2}{0.50} = 6.4.$$

2. Substitute into the ratio:

$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{6.4}\frac{1.8}{1.0}.$$

3. Compute each factor:

$$\frac{\pi^2}{6.4} = \frac{9.8696}{6.4} = 1.542, \qquad \frac{1.8}{1.0} = 1.8.$$

4. Multiply:

$$\frac{F_{I,\max}}{F_{D,\max}} = (1.542)(1.8) = 2.78.$$

Source(s): 41107_mock_exam_with_answers

O12. Question 12

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. The normalized fundamental lift frequency is $N_L = 2$. Which name is given to this flow regime?

Options:

A. Single-pair regime
B. Double-pair regime
C. Three-pair regime
D. No-separation regime

Correct answer: A. Single-pair regime

Formulario / principles used

- Normalized lift frequency:

$$N_L = \frac{f_L}{f_w}.$$

- Course convention:

$$N_L \approx 2 \Rightarrow \text{single-pair vortex shedding regime.}$$

Step-by-step solution

1. Given:
$$N_L = 2.$$
2. Match the value with the course regime map:
$$N_L = 2 \Rightarrow \text{single-pair regime.}$$
3. No further numerical calculation is required.

Source(s): 41107_mock_exam_with_answers

Froude-Krylov / diffraction / large-body questions

D1. Question 1

DIFFRACTION / LARGE BODIES. For wave diffraction around a cylinder, when is diffraction expected to be important?

Options:

A. When $D/L \gtrsim 0.2$
B. When $D/L \lesssim 0.02$
C. Only when $KC > 20$
D. Only when $Re < 1000$

Correct answer: A. When $D/L \gtrsim 0.2$

Formulario / principles used

- Diffraction parameter:
$$\frac{D}{L} = \frac{\text{characteristic body diameter}}{\text{wavelength}}.$$
- Course rule of thumb:
$$\frac{D}{L} \gtrsim O(0.2) \quad \Rightarrow \quad \text{diffraction is important.}$$
- If D/L is small, the body is slender relative to the wavelength and Morison/Froude-Krylov type loading is usually more appropriate.

Step-by-step solution

1. The question asks for the condition under which wave diffraction around a cylinder becomes important.
2. Diffraction becomes important when the cylinder is not small compared with the wavelength.
3. The course threshold is approximately:
$$\frac{D}{L} \gtrsim 0.2.$$
4. Option A states exactly this criterion.
5. Options B, C and D are traps: KC and Re are important for flow regime and viscous/separation effects, but the diffraction criterion is primarily based on D/L.

Source(s): 41107_mock_exam_with_answers / 41107_MCQ_Study_Guide

D2. Question 2

SHIP MOTION DYNAMICS - Froude-Krylov Force (Diffraction Neglected). For a ship in regular waves, under which assumption is the Froude-Krylov force dominant and diffraction can be neglected?

Options:

A. $D/L > 0.5$ (large ship/short waves)

B. $D/L < 0.2$ (small body relative to wavelength)
C. $Fn > 0.3$ (high Froude number)

Correct answer: B. $D/L < 0.2$ (small body relative to wavelength)

Formulario / principles used

- Diffraction parameter:
$$\frac{D}{L} = \frac{\text{characteristic body dimension}}{\text{wavelength}}.$$
- Small-body / Morison-type regime:
$$\frac{D}{L} < 0.2 \quad \Rightarrow \quad \text{diffraction small.}$$
- Large-body / diffraction regime:
$$\frac{D}{L} \gtrsim 0.2 \quad \Rightarrow \quad \text{diffraction may be important.}$$
- Froude-Krylov force uses the undisturbed incident-wave pressure field.

Step-by-step solution

1. The Froude-Krylov force assumes the incident wave pressure field is not strongly modified by the body.
2. This assumption is most valid when the body is small compared with the wavelength.
3. The course rule of thumb is:
$$\frac{D}{L} < 0.2.$$
4. Therefore, diffraction can be neglected and Froude-Krylov force is dominant when:
$$D/L < 0.2.$$
5. Option A is the opposite case: a large body relative to the wavelength, where diffraction matters.
6. Option C uses Froude number, which is not the diffraction criterion in this context.

Source(s): 41107_MCQ_Study_Guide / Wave-structure interactions - Regular waves (L10)

Final formula reminder for this group

Steady current around cylinders

$$Re = \frac{U_c D}{\nu}$$

$$St = \frac{f_v D}{U_c}$$

$$f_{D,\mathrm{osc}} = 2 f_v$$

$$p_s - p_0 = \frac{1}{2}\rho U_c^2$$

$$F_D = \frac{1}{2}\rho C_D D U_c^2$$

$$C_D = \frac{F_D}{\frac{1}{2}\rho U_c^2 D}$$

Oscillatory flow and wave-related cylinder questions

$$U(t) = U_m \sin(\omega t), \qquad \omega = \frac{2\pi}{T}$$

$$KC = \frac{U_m T}{D}$$

$$A = \frac{\pi D^2}{4}$$

$$F_{FK} = \rho A \dot{U}$$

$$F_{FK,\max} = \rho A U_m \omega$$

$$F_x = \frac{1}{2}\rho C_D D U|U| + \rho C_M A \dot{U}$$

$$F_I = \rho C_M A \dot{U}, \qquad F_K = \rho A \dot{U}, \qquad F_I = C_M F_K$$

$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{KC}\frac{C_M}{C_D}$$

$$N_L = \frac{f_L}{f_w}$$

$$\omega^2 = gk\tanh(kh), \qquad L = \frac{2\pi}{k}$$

$$H_{\max} \approx 0.142 L \tanh(kh)$$

Froude-Krylov and diffraction

$$F_{\mathrm{exc}} = F_{FK} + F_{\mathrm{diff}}$$

$$\frac{D}{L} < 0.2 \quad \Rightarrow \quad \text{small body, diffraction usually negligible}$$

$$\frac{D}{L} \gtrsim 0.2 \quad \Rightarrow \quad \text{large-body diffraction may be important}$$